



GAME-SPECIFIC FACULTY GUIDE

Signal House

A two-case economic escape room built around evidence, causal reasoning, and aggregate supply.

Signal House asks students to reconstruct an economic event from records scattered through a house. This guide explains how its two investigations turn those discoveries into economic reasoning and suggests ways to bring the evidence into an AD–AS discussion.

Why This Game Exists

I wanted to try an escape room where the economics itself helped students get somewhere. Escape rooms are familiar, but I hadn't seen many that worked that way. Students would search the house, find records, and gradually work out what had happened. Understanding the economics would let them move on.

I wanted them to start with pieces of the story: a household has less left over, a producer faces a higher cost, and a national report shows weaker output. They'd have to connect those records before they could explain the event as a whole.

Some of the game is ordinary searching. Then students have to compare accounts, read production records, weigh a business decision, and put the events in order. By the time they reach the AD–AS model, they have evidence to bring to it.

What Students Do

Students explore one continuous house, examining objects and recovering records. The Broken Signal opens a later investigation, The Second Harvest, which reuses the house with new evidence. Students can search broadly, revisit discoveries, and use the satchel to review retained records. Some become useful only when another discovery supplies the missing connection.

A household account may establish what happened to one household; a production record may explain one firm's decisions. Students encounter these fragments before they can make an economy-wide claim. National observations and wider reporting give the local records a broader context. Opening the exit requires students to connect the economic events, whose sequence may differ from the order of discovery.

Both investigations are untimed. Students can use increasingly explicit hints when they get stuck on an object or need to connect discoveries, and revise unsuccessful settings or arrangements as they learn more. Hint use carries no score or mastery penalty. This support keeps navigation difficulties from blocking the economics, leaving students to concentrate on connecting the evidence.



One view of the shared house. Students encounter evidence within the environment and bring records together as their investigation develops.

What the Game Is Really Teaching

In *The Broken Signal*, students find archived reports of storm damage and expensive emergency routes, then connect them to production records. A higher price alone could have several explanations. Timing, costs, firms' decisions, and national observations let them build a case for what happened and decide whether a local event had wider effects.

In the workshop, they compare the added revenue and added cost of jobs that use limited material. An already-paid, unrecoverable lease is a sunk cost. Material can stay in storage, so there's no reason to use it all on work that loses money. Students can see why a firm might cut activity even while customer orders remain available.

The production and staffing cuts then connect to national evidence of lower real output, higher unemployment, and higher prices. Higher costs across producers reduce what firms are willing to supply at a given price level. Students can use the wider evidence to explain a leftward shift in short-run aggregate supply (SRAS), moving beyond what one firm's records could establish.

Both case interpretations hold aggregate demand (AD) unchanged. Continuing customer orders help explain the first firm's cost-driven response, and the final model explicitly fixes AD. In the second case, a separate demand log covers policy and independent spending conditions. An unchanged policy setting alone would not settle what happened to other sources of demand.

The records also ask students to distinguish the measures. National output is measured at constant prices. A household can receive more nominal pay yet afford less of the same essential bundle. The price index describes a level; inflation describes its change over an interval. Mission 1 compares matching monthly inflation periods. The adjustment shown in a case does not establish an indefinitely rising or falling inflation path.

After the adverse supply shock, students face a policy choice. Tighter demand policy can ease inflation pressure while weakening output and employment further. A looser response can support work and output while adding inflation pressure. Neither repairs the disrupted input network. The illustrative policy gauges let students compare those competing objectives without reading them as calibrated forecasts.

The Second Harvest

The later case begins with a resource-saving trial: three pump-hours in place of six for the same harvest, quality, land, and other inputs. With the operating rate and other batch costs unchanged, the cost of the same batch falls from \$21 to \$15. Students reconstruct that cost from separate records.

A successful local trial alone cannot establish an aggregate supply shift. Installation records show that growers, mills, and equipment makers adopted the method across districts and a national supplier network. Students need that wider adoption alongside the cost evidence, national observations, and separate unchanged-demand condition.

The national real-output index rises from 200 to 216, while the price index falls from 100 to 96. Together, the records support a rightward SRAS shift with AD unchanged. Establishing that a beneficial method spread gives students a different problem from tracing the earlier disruption.

The comparison establishes a lower price level, without implying that prices keep falling. No corresponding labor-market series is supplied, so employment and unemployment effects remain open.

Comparing the Two Cases

Ask students what changed first in production, what justified extending the explanation to the national economy, and what supports holding AD unchanged. Use the table to compare the evidence behind each shift.

The Broken Signal	The Second Harvest
Disrupted input deliveries and costly emergency routes raise production costs.	A resource-saving method lowers the cost of the same output batch.
Reports across manufacturers and matched national indicators establish wider effects.	Adoption across producer groups and national records establish wider effects.
With AD unchanged, SRAS shifts left: real output falls and the price level rises.	With AD unchanged, SRAS shifts right: real output rises and the price level falls.

The Educational Loop

Mission 1 closes with an evidence recap, economic explanation, labeled AD–AS graph, and transfer forecast. Students can revise their forecast after feedback before opening the results screen. Mission 2 concludes with its evidence-based explanation and comparison of the cases; the explicit graph and transfer forecast belong to Mission 1.

Experience → Consequence → Economic Explanation → Formalization → Transfer

Bringing in the AD–AS Model

1. Ask what students thought was happening before opening the exit and which records changed or confirmed their interpretation.
2. Have them build a causal account, separating local observations from evidence of national change.
3. Draw AD and SRAS with real output on the horizontal axis and the price level on the vertical axis. State the unchanged-AD condition.

4. Map The Broken Signal to SRAS left. Ask students to justify the output and price-level changes using their records.
5. Map The Second Harvest to SRAS right. Ask which evidence supports lower costs and wider adoption.
6. Compare the evidence supporting each interpretation, including unchanged demand, and ask what would make students reconsider.

Using It in Class

These are optional ways to use the investigation; choose what fits your course.

Before formal AD–AS instruction

Let students investigate and explain what the records suggest in everyday language. They can reconstruct the mechanism before knowing its formal name. After play, use their evidence to introduce the graph.

During AD–AS instruction

Use The Broken Signal after introducing supply-side shifts. Students can pause when the first door opens to construct the graph together before its reveal. The Second Harvest then supplies a contrasting case.

After instruction

Use either case, subject to the second mission's unlock requirement, as evidence-based practice. Ask students to defend the supply shift with records from the investigation and explain what rules out a competing interpretation.

Independent/student review

Students can revisit evidence and work through Mission 1's graph and forecast. Mission 2 unlocks at the first exit, independently of the forecast, so invite students to complete that follow-up. You can request a reflection or screenshot through your usual course workflow; the game sends no faculty completion report.

Questions to Consider

Select any of these prompts for discussion or a short reflection.

1. Which record first made you suspect a production-side change? What other explanation was still possible at that point?
2. What additional evidence did you need before treating a local cost or production change as economy-wide? Compare the two cases.
3. In The Broken Signal, how did the cost records explain firms' decisions while customer orders remained available? Which costs mattered to accepting additional work?
4. Why did the first case create a difficult policy choice? What could the policy control change, and what remained unresolved?
5. In The Second Harvest, what did holding harvest, quality, and other inputs constant let you infer from fewer pump-hours?
6. Why was the local trial insufficient for an aggregate conclusion? Which evidence established that other producers had adopted the improvement?
7. What supports holding AD unchanged in each case? Distinguish evidence about one firm's orders from the broader demand condition used in the model.
8. A third case reports rising output and a rising price level. What additional evidence would you request before deciding which curve shifted?

Common Misconceptions

- “Higher prices mean demand increased.” Ask students to explain the cost and output evidence alongside the price change.
- “One firm's cost change proves aggregate supply shifted.” Look for evidence of wider exposure or adoption and national consequences.
- “The rise in unemployment caused the supply shock.” Revisit the timing: the first case links the input disruption to costs, production cuts, and weaker employment.
- “The policy machine has one setting that fixes everything.” Ask students to describe what happened to both objectives in each trial and whether the input network was repaired.
- “A lower price level means continuing deflation.” A comparison between two price levels establishes a change over that interval; its continuation needs further evidence.
- “A productive local trial proves economy-wide growth.” Separate the trial result from adoption and national output evidence. The second case establishes a short-run supply improvement, without measuring a permanent growth rate.